

Lecture 1: Review of Statistical Concepts

In this lecture, we will review some basic statistical concepts that we will use in the rest of this course. Most of these are concepts that you should have seen before in previous classes, although it is okay if some of them are new.

Probability

A **random variable** is a variable whose value depends on the random realization of events. For example, if we do not know what the return on the stock market will be this year, then we can say that R is a random variable representing the annual return on the stock market. Similarly, we might imagine selecting an American adult at random, and measuring their height. Their height in inches, H , would be a random variable, because although we might know the distribution of heights in the population, we do not know which person will be selected.

Each random variable has a **distribution**, which reflects the probability of taking on different values. This distribution can be expressed with the **cumulative distribution function (cdf)**. The cdf, $F(\cdot)$, of a random variable, X , is the probability that the random variable is less than or equal to a given value: $F(x) = P(X \leq x)$. For continuous random variables, we also often use the **probability density function (pdf)** to characterize the distribution. The pdf is just the derivative of the cdf. For discrete random variables, we instead can use the **probability mass function (pmf)**. The pmf is the probability that the variable takes on a particular value; e.g. the pmf for X at 5 is $P(X = 5)$. Note that the pdf must integrate to the cdf, $\int_{-\infty}^a f(x) dx = F(a)$. Similarly, for a discrete distribution, the pmf must add up to the cdf $F(a) = \sum_{x \leq a} P(x)$.

Exercise 1. Explain why the cdf must be weakly increasing (it can never decrease).

Exercise 2. Explain why $F(x) \rightarrow 1$ as $x \rightarrow \infty$.

We often have more than one random variable. For example, consider the random variables X and Y . We can consider the **joint distribution** of these variables using the same tools. For example, the cdf is now the probability that $X \leq x$ and $Y \leq y$, which we write as $F(x, y) = P(X \leq x, Y \leq y)$. The pdf is the derivative of the cdf with respect to both x and y .

Mean, Variance, and Covariance

Rather than thinking about the full distribution, we often want to have one or two numbers that summarize the distribution. One such measure is the **mean**, also referred to or **expected value**. The expected value of X is denoted $\mathbb{E}[X]$. We can compute the expectation as $\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x) dx$ for a continuous variable, or as $\mathbb{E}[X] = \sum_x x \cdot P(x)$.

Another important measure is the **variance**. The variance is defined as the average squared deviation from the mean. That is, $\text{Var}(X) := \mathbb{E}[(X - \bar{X})^2]$, where $\bar{X} = \mathbb{E}[X]$. We also often refer to the **standard deviation**, which is the square root of the variance.

In this course, we will be especially interested in how the distribution of one variable is related to the value of other variables. One very useful concept here is the **covariance**. The covariance of X and Y is defined as $\text{Cov}(X, Y) = \mathbb{E}[(X - \bar{X})(Y - \bar{Y})]$. The covariance measures the linear relationship between X and Y .

Closely related to the covariance is the **correlation**, sometimes referred to as the **correlation coefficient**. The correlation is defined as $\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \cdot \text{Var}(Y)}}$. The correlation coefficient is convenient because it is scale-invariant: it does not change even if we change the units of X or Y (see exercise below).

We say that two variables are **uncorrelated** if their correlation is zero (equivalently, if their covariance is zero). A stronger condition is to say that two variables are **independent**. Two variables, X and Y are

independent if the cdf for (X, Y) can be written as a product of the cdfs for X and Y ; that is, X and Y are independent if $F_{X,Y}(x, y) = F_X(x) \cdot F_Y(y)$ for all values of x and y . For continuous variables, this is equivalent to saying that $f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y)$. For discrete variables, it is equivalent to saying that $P(X = x, Y = y) = P(X = x) \cdot P(Y = y)$. Note that independence implies uncorrelatedness, but uncorrelatedness does not necessarily imply independence.

Exercise 3. Show that $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

Exercise 4. Show that $\text{Var}(X) = \text{Cov}(X, X)$.

Properties of the Mean, Variance, and Covariance The mean, variance, and covariance have a number of important special properties. In what follows, we will assume that lower case letters (a, b, c , and d) are constants, and uppercase letters (X, Y , and Z) are random variables. First, we have the following properties for the expected value:

- $\mathbb{E}[a + b \cdot X] = a + b \cdot \mathbb{E}[X]$
- $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$
- $\mathbb{E}[a \cdot X + b \cdot Y + c] = a \cdot \mathbb{E}[X] + b \cdot \mathbb{E}[Y] + c$

Together, these properties are referred to as the **linearity of expectation**.

These properties also imply corresponding properties for the variance:

- $\text{Var}(a + b \cdot X) = b^2 \cdot \text{Var}(X)$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \cdot \text{Cov}(X, Y)$

Finally, note the following properties of the covariance:

- $\text{Var}(X) = \text{Cov}(X, X)$
- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- $\text{Cov}(X, a) = 0$
- $\text{Cov}(X + Y, Z) = \text{Cov}(X, Z) + \text{Cov}(Y, Z)$
- $\text{Cov}(a + b \cdot X, c + d \cdot Y) = b \cdot d \cdot \text{Cov}(X, Y)$

We will use these properties extensively in this course; I encourage you to memorize them.

For each of the exercises below, use the linearity of expectation to prove the corresponding properties about variance, covariance, or correlation.

Exercise 5. Let a and b be constants. Show that $\text{Var}(a + b \cdot X) = b^2 \cdot \text{Var}(X)$.

Exercise 6. Let a, b, c and d be constants. Show that $\text{Cov}(a + b \cdot X, c + d \cdot Y) = b \cdot d \cdot \text{Cov}(X, Y)$.

Exercise 7. Let a, b, c and d be constants. Show that $\text{Corr}(a + b \cdot X, c + d \cdot Y) = \text{Corr}(X, Y)$.

Conditional Expectation and Variance

We can also think about the **conditional expectation** and **conditional variance**. For $\mathbb{E}[Y | X]$, this is the expectation of Y given the value of X , and for $\text{Var}(Y | X)$, this is the variance of Y given the value of X . We can compute each of these values in the same way as we did the unconditional expectation, but using the conditional distribution rather than the unconditional distribution. Note that these are very important concepts for this course: most of this course will be about modeling and estimating $\mathbb{E}[Y | X]$.

We have two very convenient properties when it comes to the conditional expectation and variance. The first is the **law of iterated expectations**:

$$\mathbb{E} [\mathbb{E} [Y | X]] = \mathbb{E} [Y]$$

This law states that the average value of Y is equal to the average of the conditional averages.

Exercise 8. The average height for adults in the United States is 68.9 inches for men, and 63.5 inches for women. Suppose that the number of men and women is equal. What is the average height of all adults in the United States?

Exercise 9. Suppose that $\mathbb{E} [X] = \mu$, $\text{Var} (X) = \sigma^2$, and $\mathbb{E} [Y | X] = aX^2 + bX + c$. Solve for $\mathbb{E} [Y]$.

The second is the **law of total variance**:

$$\text{Var} (Y) = \text{Var} (\mathbb{E} [Y | X]) + \mathbb{E} [\text{Var} (Y | X)]$$

This law states that the variance of Y is equal to the variance of the conditional expectation plus the expectation of the conditional variance.

One more useful property is that when we are conditioning on something, we can treat it as a constant and pull it out of the conditional expectation. Let $f(X)$ be some function of X :

$$\mathbb{E} [f(X) \cdot Y | X] = f(X) \cdot \mathbb{E} [Y | X]$$

The Normal Distribution

Finally, although there are many different distributions, the most important is the **normal distribution**, also known as the **Gaussian distribution**. This distribution is defined by two parameters, the mean and the variance. To say that X follows a normal distribution with mean μ and standard deviation σ , we write $X \sim N(\mu, \sigma^2)$. We refer to the distribution $N(0, 1)$ as the standard normal distribution. The univariate normal distribution is defined by the pdf

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}.$$

We sometimes use $\phi(x)$ to refer to the pdf of the standard normal, and $\Phi(x)$ to refer to the cdf of the standard normal.

The normal cdf does not have an analytic solution: instead, it must be evaluated on the computer. Given the importance of the normal distribution, the standard normal cdf is readily available in statistical and other software. To evaluate the cdf for other normal distributions, we can standardize our variables: subtracting the mean and dividing by the standard deviation. For the normal distribution, it is true that $F\left(\frac{x-\mu}{\sigma}\right) = \Phi(x)$.

Beyond the univariate normal, there is also a corresponding **multivariate normal distribution**. We denote the multivariate normal with $N(\mu, \Sigma)$, where μ is a vector of means, and Σ is the variance-covariance matrix. We say that two or more variables are distributed **multivariate normal** or **jointly normal** if they are distributed according to a multivariate normal distribution. The multivariate normal has an extremely convenient property: if X and Y are jointly normal, then $a + bX + cY$ is also normally distributed, for any constants a, b , and c . This property naturally extends to any number of variables, as long as they are all jointly normal. Note that it is not sufficient to say that X is normally distributed and Y is normally distributed, it must be the case that they are jointly normal for this property to hold.

Exercise 10. Suppose that male height in inches, Y , is distributed $Y \sim N(68.9, 2.8^2)$, and that female height in inches, X , is distributed $X \sim N(63.5, 2.5^2)$. What is the probability that a randomly selected man is shorter than a randomly selected woman? Express your answer in terms of the standard normal cdf $\Phi(\cdot)$. *Hint: Re-write the probability $P(X > Y)$ as $P(X - Y > 0)$.*

Exercise 11. Why can it not actually be true that height is normally distributed?

Statistics

Statistics is the inverse of probability. In probability, we assume that a variable has a given distribution, and ask about the outcomes that might be generated by that distribution. In statistics, we observe data, and ask what distribution might have generated that data. We call this distribution the **data-generating process (DGP)**. We draw an important distinction between the **sample**, which is the draw of data that we observe, and the **population**, which reflects the underlying data-generating process.

Estimators

One of the essential concepts in statistics is an **estimator**. In principle, an estimator can be any function of the data. But in practice, an estimator is intended to provide a “good” estimate of some parameter of the data-generating process. There are many notions of what makes a good estimator, but one important property is **unbiasedness**. Let X denote our sample data, and let $\hat{\theta}(X)$ denote an estimator of the parameter θ . We say that $\hat{\theta}$ is unbiased if $\mathbb{E}[\hat{\theta}(X)] = \theta$. More generally, we refer to the bias of $\hat{\theta}$ as $\mathbb{E}[\hat{\theta}(X)] - \theta$.

Exercise 12. Suppose that X_i has mean μ for all i . Let $\bar{X}_n$ denote the sample mean of X_i , in a sample of size n . Show that $\bar{X}_n$ is unbiased.

Another important characteristic of an estimator is the **sampling distribution**. The sampling distribution is the distribution of the estimator across different draws of the sample. Aside from the mean of the sampling distribution, which determines the bias, another important property of the sampling distribution is the **standard error**. The standard error is the standard deviation of the estimator’s sampling distribution.

In order to determine the sampling distribution, we often require assumptions about how the data were drawn. In particular, we often assume that each observation is **independent and identically distributed (iid)**. This means that not only is each observation independent from other observations, it is also drawn from the same distribution. With the assumption of iid observations, we can often derive properties of the sampling distribution, especially as the number of observations grows large. We will explore this further in the next lecture.

Exercise 13. Suppose that X_i is distributed iid, with $\text{Var}(X_i) = \sigma^2$ for all i . Let $\bar{X}_n$ denote the sample mean of X_i , in a sample of size n . Show that $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$.

Hypothesis Tests and Confidence Intervals

In statistics, we often wish not just to develop estimators, but also to do inference: we want to know if it is plausible that a parameter takes a particular value, or lies in some set. To do this, the most common approach is to conduct a **hypothesis test**. A hypothesis test has a few components:

- A **null hypothesis**, H_0 . This is the hypothesis that we are testing about the underlying parameters. Note that although we often also have an alternative hypothesis, H_1 , this may be as simple as “the null is false.”

- A **test statistic**, $\tau(X)$, which is a function of the data, X . Ideally, the test statistic will be low if the data appears consistent with the null, and high if the data does not appear consistent with the null.
- A **critical value**, c , above which the hypothesis test will reject the null.

Hypothesis tests typically can be expressed as a rule:

$$\text{Reject if and only if } \tau(X) > c$$

Note that hypothesis tests can either reject the null hypothesis or “fail to reject” the null hypothesis: we never “accept” the null. Just because the hypothesis test cannot reject the null does not mean that other hypotheses are not also possible.

For example, our hypothesis test might be a test to see if a patient has a disease: our null is that the patient is healthy, but if the blood sample has enough antigens, then we will reject the null in favor of the alternative hypothesis (they are sick). This medical example gives us the language of “positive” and “negative,” which we use to describe the possible outcomes of the hypothesis test. There are four possible outcomes: we can reject a false null (true positive), we can reject a true null (**false positive**), we can not reject a false null (**false negative**), or we can not reject a true null (true negative).¹

The **size** of a test is the probability that, conditional on the null being true, the test rejects (a false positive). We will use the term size interchangeably with the **level** or **significance level**.² A related but distinct concept is the **p-value**, which is the lowest possible size for which the data will still reject the null (we find this by setting the critical value equal to the value of the test statistic in the data, and then figuring out the size associated with that critical value). An equivalent way to think of the p-value is to say that is the probability, under the null, that we would observe a test statistic that is at least as large as the one we observed.

In practice, the most common test statistic takes the form

$$\tau(X; \beta_0) = \frac{|\hat{\beta} - \beta_0|}{\text{Standard Error}}$$

where β_0 is our null hypothesis about β , $\hat{\beta}$ is our estimator, and we know that $\frac{\hat{\beta} - \beta_0}{\text{Standard Error}}$ is distributed according to a standard normal under the null (usually this is an asymptotic result). This test statistic is called the **two-sided t-test**. In this case, under the null we have that $\tau(X; \beta_0) \sim |N(0, 1)|$, and can use this to generate the critical value that yields our desired size. For example, to achieve a size of 5%, we use a critical value of approximately 1.96.

A related idea to hypothesis testing is the **confidence set**. An $X\%$ confidence set that has the following property: the true value of β will be in the confidence set $X\%$ of the time. If β is one-dimensional and the confidence set takes the form of an interval $[\beta_{\text{Low}}, \beta_{\text{High}}]$, then we call it a **confidence interval**. A convenient fact is that if we have hypothesis tests for each null hypothesis β_0 , then we can construct a confidence set by “inverting” the hypothesis test. For example, if we want a 95% confidence set, then we can test every possible value of β using a 5%-sized test, and include in the confidence set all of the points where the test failed to reject the null. The probability of this procedure excluding the true value of β is 5%, so the probability that it includes the true value is 95%.

Exercise 14. Suppose that I wish to construct a 95% confidence set for the population mean. To do this, I draw my data, $\{X_i\}_{i=1}^N$, and then I roll a (fair) twenty-sided die. If I roll a 20, I return an empty confidence set, otherwise I return the entire real line. Is this a valid 95% confidence set? Is it a useful one?

¹A false positive is also called a type I error, and a false negative is also called a type II error. I have never met anyone who can remember which one is which, without looking it up.

²Technically, a test has significance level α if the size is less than or equal to α . But in practice, we always state the lowest possible significance level we can, which is just the size of the test.

For the two-sided t-test, this yields easy to compute confidence intervals. Suppose we wish to construct 95% confidence intervals. The 5% test fails to reject if and only if $\frac{|\hat{\beta} - \beta_0|}{\text{Standard Error}} \leq 1.96$. Rearranging terms, this yields $|\hat{\beta} - \beta_0| \leq 1.96 \times \text{SE}$, which becomes the interval: β_0 is in $[\hat{\beta} - 1.96 \times \text{SE}, \hat{\beta} + 1.96 \times \text{SE}]$.

Finally, to clear up confusion about confidence sets, it is helpful to define its Bayesian counterpart, the **credible interval/set**. You do not need to know what a credible set is for this course, but is useful because it helps us better understand what a confidence set is. A 95% credible set is a set of points such that, after seeing the data, you believe that there is a 95% probability that the true value of β lies within the credible set. The key distinction between a 95% confidence set and a 95% credible set has to do with *ex ante* vs. *ex post*. I can tell you, before seeing the data (*ex ante*), that the true value will be in the confidence set 95% of the time, while for a credible set, I can tell you after seeing the data (*ex post*) that the true value is in the credible set with 95% probability. The latter is not true for confidence sets. For example, suppose I have very strong reasons to believe that $\beta > 0$, but my confidence set returns $[-5, -1]$. *Ex ante*, there was only a 5% probability that the confidence set would not contain the true value, but *ex post* there was good reason to believe that this was one of the 5% of times that the confidence set made a mistake.